Affect Detection in Texts

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Applications

- Sentiment Analysis
  - Text categorization according to affective relevance
  - Opinio exploration for market analysis

- Computer Assisted Creativity
  - Automatic personalized advertising and persuasive communication.

- Verbal expressivity in HCI
  - Affective word selection and understanding are crucial for realizing appropriate and expressive conversations.
  - Question answering systems

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Emotion through typography

- Typographers are attuned to subtle features while designing

Aa

GEORGIA
designed by Matthew Carter

Aa

VERDANA
designed by Matthew Carter

Typeface

minimum

Johann Herder first proclaimed in 1772 that the basis of a nation was a language with its oral, traditional songs and stories. If there is a language, then it must be written down, given an alphabet and

minimum

standardized by deliberate selection from all its local variants. A dictionary must be written, and grammar must be provided for the children. A history of the people must be compiled. Folk-tales

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and poetry must be collected and published to lay the base for a modern culture – or for a 'national intelligentsia' which will go on to compose a national literature.

Symmetry

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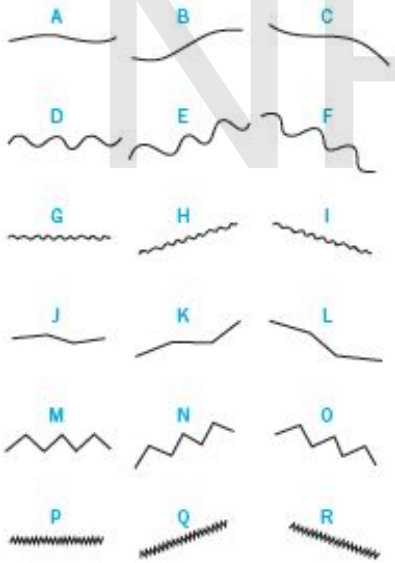
Poffenberger, & Barrows (1924):

- A line going downwards was shown to make us feel “doleful,” while a “joyous” like takes our eyes upwards.

Simple lines can convey emotions



Emotion	Feeling value of lines	Suggested typeface equivalent	
Angry, agitating, furious	Angles, sloping forwards	angry angry	*Cinema* italic *Klute*
Harsh	Sharp angles, either direction	harsh harsh	*Klute* Modified *Klute*
Playful, merry	Tighter curves, sloping forwards	playful	Modified *Bodoni Poster* italic
Sad, weak	Gentle curves, sloping down	sad	Modified *Cooper Black*
Happy, friendly	Gentle curves, balanced	happy	*Cooper Black*
Calm, gentle	Gentle curves, balanced	calm	*VAG Rounded*

Juni, S., & Gross, J. S. (2008)

- Articles received ratings as being funnier and angrier, in other words more satirical, when it was read in Times New Roman in comparison to Arial.

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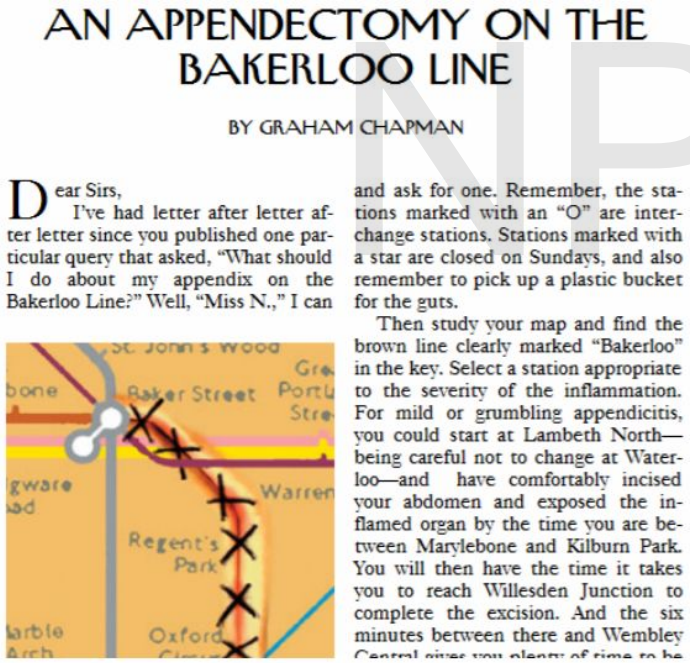
The Aesthetics of Reading

- Good typography does elevate mood (Larson, K., & Picard, R., 2005)



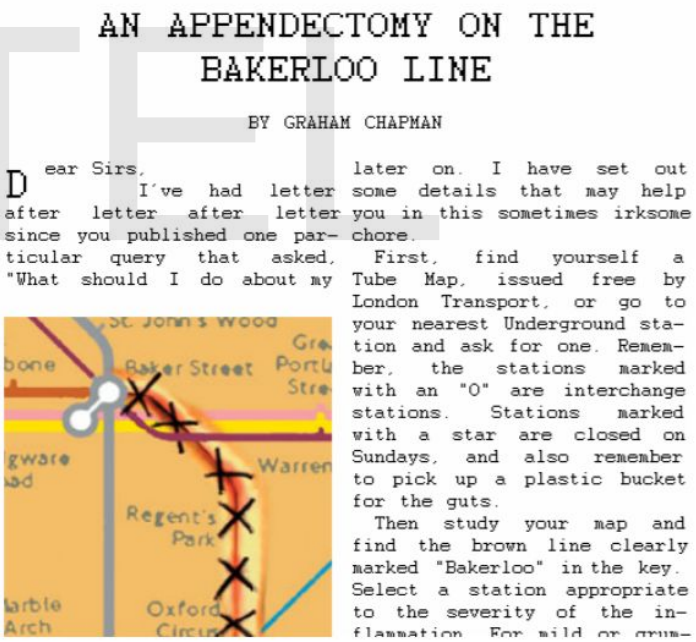
AN APPENDECTOMY ON THE BAKERLOO LINE

BY GRAHAM CHAPMAN

Dear Sirs,

I've had letter after letter after letter since you published one particular query that asked, "What should I do about my appendix on the Bakerloo Line?" Well, "Miss N.," I can and ask for one. Remember, the stations marked with an "O" are interchange stations. Stations marked with a star are closed on Sundays, and also remember to pick up a plastic bucket for the guts.

Then study your map and find the brown line clearly marked "Bakerloo" in the key. Select a station appropriate to the severity of the inflammation. For mild or grumbling appendicitis, you could start at Lambeth North—being careful not to change at Waterloo—and have comfortably incised your abdomen and exposed the inflamed organ by the time you are between Marylebone and Kilburn Park. You will then have the time it takes you to reach Willesden Junction to complete the excision. And the six minutes between there and Wembley Central gives you plenty of time to be



AN APPENDECTOMY ON THE BAKERLOO LINE

BY GRAHAM CHAPMAN

Dear Sirs,

I've had letter after letter after letter since you published one particular query that asked, "What should I do about my

later on. I have set out some details that may help you in this sometimes irksome chore.

First, find yourself a Tube Map, issued free by London Transport, or go to your nearest Underground station and ask for one. Remember, the stations marked with an "O" are interchange stations. Stations marked with a star are closed on Sundays, and also remember to pick up a plastic bucket for the guts.

Then study your map and find the brown line clearly marked "Bakerloo" in the key. Select a station appropriate to the severity of the inflammation. For mild or grum-

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The Aesthetics of Reading

- Relative subjective duration (RSD)
  - Participant's perception of how long they have been performing a task.
  - Poor typography condition underestimated by 24 seconds on average, while good typography condition underestimated by 3 minutes and 18 seconds on average.
  - Good quality typography is responsible for greater engagement during the reading task.
- Performance in the candle task
  - 4 of 10 participants in the good typography
  - condition while 0 of 9 participants in the poor
  - typography condition.
  - Better mood in good typography conditions.

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Negative/positive is not enough

- Positive/Negative is an active area but fine-grained emotion annotation could increase the effectiveness.
- For example, the two emotions Fear and Anger both express negative opinion of a person toward something, but the latter is more relevant in marketing or socio-political monitoring of the public sentiment.
- It has been shown that fearful people tend to have pessimistic view of the future, while angry people tend to have more optimistic view (Lerner and Keltner, 2000).
- Moreover, fear generally is a passive emotion, while anger is more likely to lead to action (Miller et al., 2009).

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Categorical/Dimensional?

- The dimensional models have been used very scarcely in the emotion detection literature, but shown to be promising as a model to represent emotions in textual data [Calvo and Mac Kim, 2013].

- For example, it is essential to identify the difference between Fear and Anger
- Fear
  - Valence: Negative,Arousal: Low/High , Dominance: Submissive
- Anger
  - Valence: Negative,Arousal: Low/High , Dominance: Submissive

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Complexity of Emotions

- "The cook was frightened when he heard the order, and said to Cat-skin, You must have let a hair fall into the soup; if it be so, you will have a good beating."
  - – which expresses fear.

- "When therefore she came to the castle gate she saw him, and cried aloud for joy."
  - – which is the expression for joy.

- "Gretel was not idle; she ran screaming to her master, and cried: You have invited a fine guest!"–
  - which is the expression for angry-disgusted.

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Complexity of Emotions: Implicit

- Emotion expression is very context sensitive and complex.
- Considerable portion of emotion expressions are not explicit [Lee, 2015].
  - ’be laid off’ or ’go on a first date’ which contains emotional information without specifying any emotional lexicon.
- Cambria et al. (2009) proposed an approach to overcome this issue by building a knowledge base that merges Common Sense and affective knowledge. E.g.
  - Spending time with friends causes happiness.
  - Getting into a car wreck makes one angry.

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Complexity of Emotions: Metaphors

- Expressions of many emotions, such as anger are metaphorical, thus could not be assessed by the literal meaning of the expression (e.g. 'he lost his cool' or 'you make my blood boil').
- Difficult to create a lexical, or machine learning method to identify emotions in text, without first solving the problem of understanding of metaphorical expressions.
- Emotions have a very complex conceptual structure, and this structure could be studied by systematic investigation of expression that are understood metaphorically. [Lakoff, 2008]

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Complexity of Emotions

Problem of detecting emotions from text is more than a multi-class classification problem

- Complexity of human emotions,
- Implicit expressions,
- Frequent use of metaphors,
- Importance of context in identifying emotions,
- Cross cultural and intra-cultural variations of emotions.
- ???

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Databases

- ISEAR, International Survey On Emotion Antecedents And Reactions (Scherer and Wallbott, 1994)
  - 3000 people around the world who were asked to report situations in which they experienced each of the seven major emotions (joy, fear, anger, sadness, disgust, shame, and guilt), and how they reacted to them.
- EmotiNet knowledge base
  - They started from around a thousand samples from the ISEAR database and clustered the examples within each emotion category based on language similarity.
- Alm's annotated fairy tale dataset (Alm et al., 2005),
  - Consisting of 1580 sentences from children fairy tales, also annotated with six Ekman's emotions.
- SemEval-2007 (Strapparava and Mihalcea, 2007),
  - Consists of 1250 news headlines extracted from news websites, and annotated with six Ekman's emotions.

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Affective Lexical Resources

- Affective Norms for English Words (ANEW)
  - (Bradley and Lang, 1999)
  - 2000 words which has been annotated based on dimensional model of emotions, with three dimensions of valance, arousal and dominance

Description	Word No.	Valence Mean(SD)	Arousal Mean(SD)	Dominance Mean (SD)	Word Frequency
abduction	621	2.76 (2.06)	5.53 (2.43)	3.49 (2.38)	1
abortion	622	3.50 (2.30)	5.39 (2.80)	4.59 (2.54)	6
absurd	623	4.26 (1.82)	4.36 (2.20)	4.73 (1.72)	17
abundance	624	6.59 (2.01)	5.51 (2.63)	5.80 (2.16)	13
abuse	1	1.80 (1.23)	6.83 (2.70)	3.69 (2.94)	18
acceptance	625	7.98 (1.42)	5.40 (2.70)	6.64 (1.91)	49
accident	2	2.05 (1.19)	6.26 (2.87)	3.76 (2.22)	33

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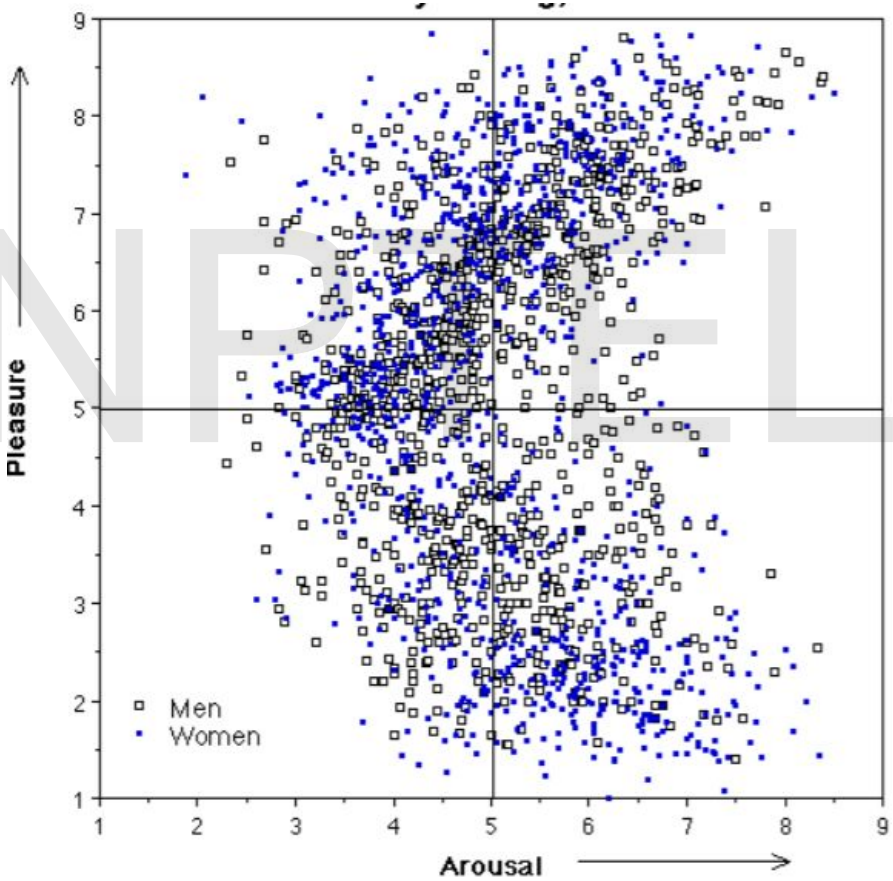
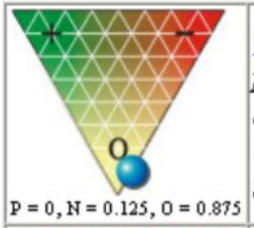
Scatter Plot: ANEW

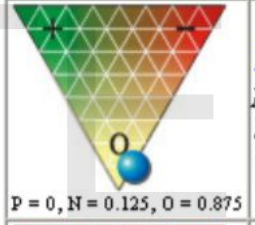


SentiWordNet

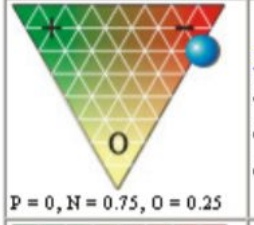
- SentiWordNet
  - (Esuli and Sebastiani, 2006; Baccianella et al., 2010)
  - Lexical resource that focuses on polarity of subjective terms
    - Opinion mining (OM)
  - *Obj(s) = 1 - [Pos(s) + Neg(s)]*
  - Rationale: The given text has factual nature, if there is no presence of a positive or a negative opinion on it.



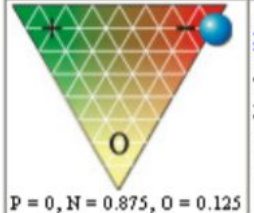
short(1)
primarily temporal sense; indicating or being or seeming to be limited in duration; "a short life"; "a short flight"; "a short holiday"; "a short story"; "only a few short months"



short(2)
primarily spatial sense; having little length or lacking in length; "short skirts"; "short hair"; "the board was a foot short"; "a short toss"



short(3)
low in stature; not tall; "his was short and stocky"; "short in stature"; "a short smokestack"



inadequate(2) poor(7) short(4)
not sufficient to meet a need; "an inadequate income"; "a poor salary"; "money is short"; "on short rations"; "food is in short supply"; "short on experience"

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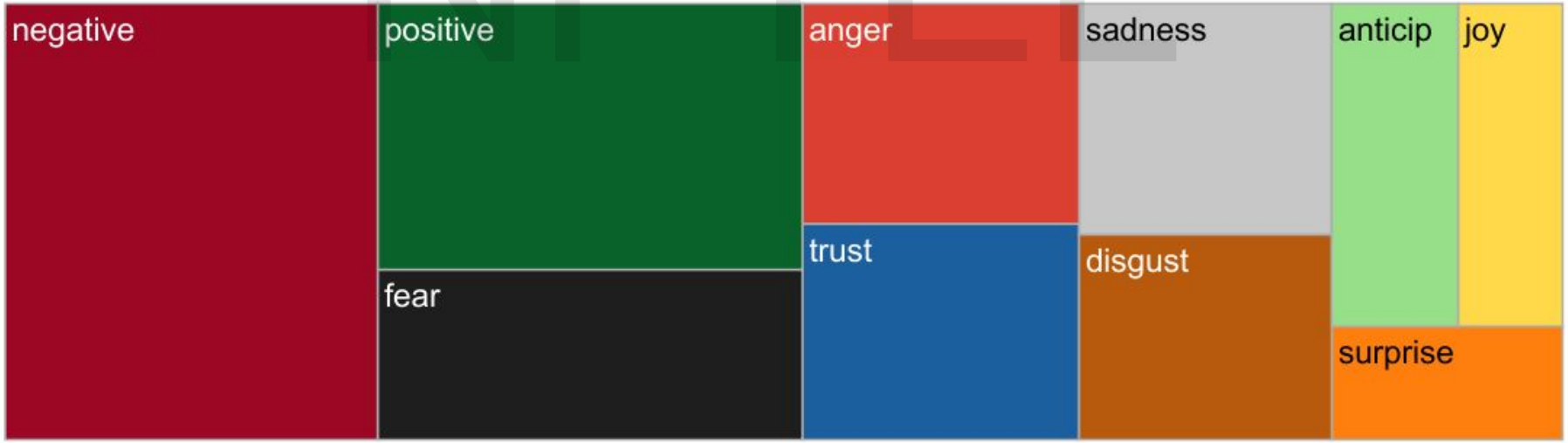
NRC

- NRC Word-Emotion Association Lexicon
  - (Mohammad, S. M., & Turney, P. D. (2013)
  - Prior work largely focused on positive and negative sentiment.
  - Crowd-sourcing to annotate around 14000 words in English language.
  - Versions of the lexicon in over one hundred languages by translating the English terms using Google Translate.



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Other Lexical Resources

- Linguistic Inquiry and Word Count (LIWC)
  - (Pennebaker et al., 2001)
  - Consisting 6400 words annotated for emotions
  - Each word or word stem defines one or more word categories.
  - >70 classes
  - E.g. the word 'cried' is part of four word categories: sadness, negative emotion, overall affect, and a past tense verb.
- WordNet Affect
  - Strapparava, C., & Valitutti, A. (2004).
  - Developed starting from WORDNET, through a selection and tagging of a subset of synsets representing the affective meanings.
- DepecheMood.
  - Staiano, J., & Guerini, M. (2014).
  - Crowd-sourcing to annotate thirty five thousands words.

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Recognizing: Unsupervised

- For categorical models, NRC can be used as the lexicon.
- For dimensional models, ANEW can be used as lexicon.
- Then the emotion of the text can be assigned based on closeness (cosine similarity) of its vector to the vectors for each category or dimension of emotions (Kim et. al, 2010).
- If we define the similarity between a given input text, I, and an emotional class, E_j, as sim(I, E_j), the categorical classification result, CCR, is more formally represented as follows:

$$\mathrm{CCR}(I) = \begin{cases} \arg\max_{j} \left(\mathrm{sim}(I, E_j)\right) & \text{if } \mathrm{sim}(I, E_j) \geq t \\ \text{“}neutral\text{”} & \text{if } \mathrm{sim}(\mathrm{I}, \mathrm{E_j}) < \mathrm{t} \end{cases}$$

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